

UDSQL - Database Management System

Introduction

This document provides an overview of the UDSQL project, a simple file-based database management system implemented in Python. It describes the functionality of the three main classes: Main, Database, and Table.

1. Main Class

The Main class is responsible for handling user queries through a command-line interface. It allows users to create, select, and drop databases, as well as execute SQL-like queries.

Methods:

- `run()`: Handles user input and processes queries.

2. Database Class

The Database class manages the database operations, including creating tables, inserting records, updating records, deleting records, and executing queries.

Key Features:

- Creates and manages tables.
- Stores metadata using the pickle module.
- Supports INSERT, UPDATE, DELETE, SELECT, and DROP TABLE.

Methods:

- `__init__(name)`: Initializes a new or existing database.
- `create_table(table_name, primary_key, columns)`: Creates a new table.
- `_save_metadata()`: Saves database metadata.
- `_load_metadata()`: Loads database metadata.
- `execute_query(query)`: Parses and executes user queries.

3. Table Class

The Table class handles individual table operations such as inserting, updating, deleting, and selecting records.

Key Features:

- Stores data in text files.
- Supports primary keys for uniqueness.
- Reads and writes records efficiently.

Methods:

- `__init__(db_path, name, primary_key, columns)`: Initializes a table.
- `_save_to_file()`: Saves records to a file.
- `load_from_file()`: Loads records from a file.
- `insert(record)`: Inserts a new record.
- `update(column, new_value)`: Updates all records in a column.
- `update_where(column, new_value, key, key_value)`: Updates records based on a condition.
- `delete()`: Deletes all records.
- `delete_where(key, key_value)`: Deletes records based on a condition.
- `drop()`: Deletes the table file.
- `select_where(column, operator, key, key_value)`: Allows to retrieve data from the table using a condition.

Possible Operations

Here are some example queries that users can do in the application:

Create a Database:

```
CREATE DATABASE UDsql
```

Create a Table:

```
CREATE TABLE user (id int,name varchar(255),age int) # id will be the table's primary_key
```

Insert a new record:

```
INSERT INTO user (id,name,age) VALUES (1,'pepe',20)
```

Update a record:

```
UPDATE user SET name = 'pepa' WHERE age = 20
```

Select something:

```
SELECT name FROM user WHERE id = 1
```

Drop table:

```
DROP TABLE user
```

Drop database:

```
DROP DATABASE UDsql
```

Conclusion

This document outlines the basic structure and functionality of the UDSQL project. The system provides a simple yet powerful way to handle database-like operations using Python file handling techniques.